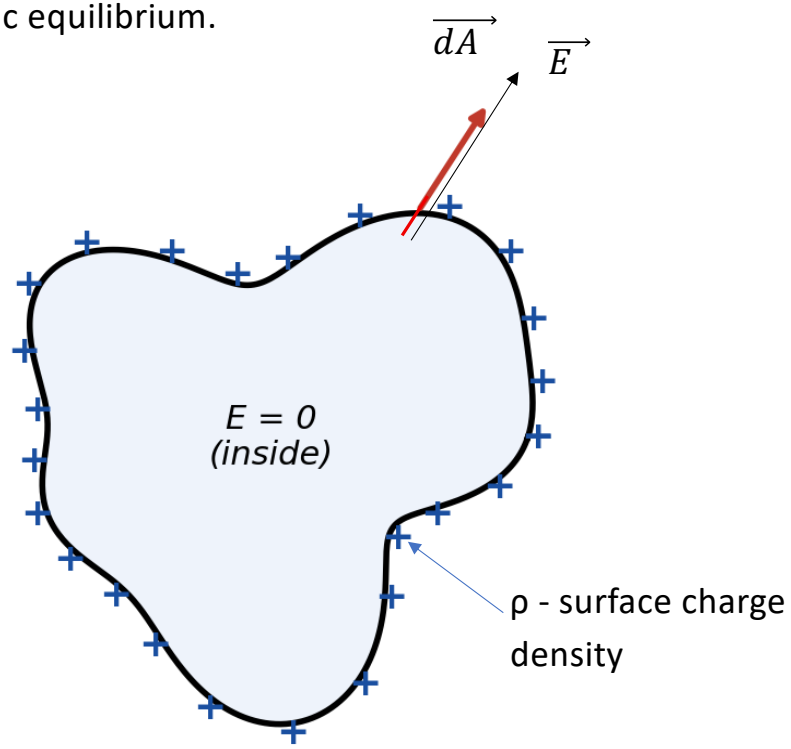
PHY1203
Applications in Electrostatics
August 14, 2026

Conductors at Electrostatic Equilibrium

- Conductors contain electrons, which are free to move about within the materials. When there is no net motion of charge, the conductor is in electrostatic equilibrium.
- Example: A Charged Conductor
- Electric charge resides on the surface.
- E (electric field) inside the conductor is zero.



- E just outside the conductor can be obtained as follows:
- From Gauss's theorem:

$$\text{Øtot} = \Sigma Q / \varepsilon_o$$

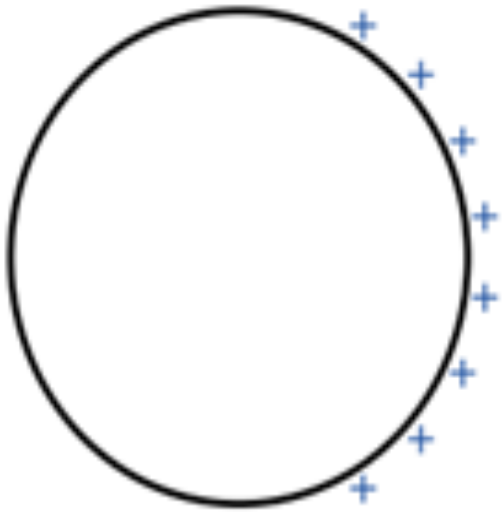
$$\vec{E} \cdot \vec{dA} = \rho \cdot dA / \varepsilon_o$$

$$E \times dA = \rho \times dA / \varepsilon_o$$

$$E = \rho / \varepsilon_o$$

- For an irregularly shaped conductor, the charge density ρ is proportional to the curvature, which is proportional to 1/r, where r is the radius of curvature.
- At sharp edges, r is very small, meaning there is a very large charge density.
- Therefore, at sharp edges the electric field E is also very high.

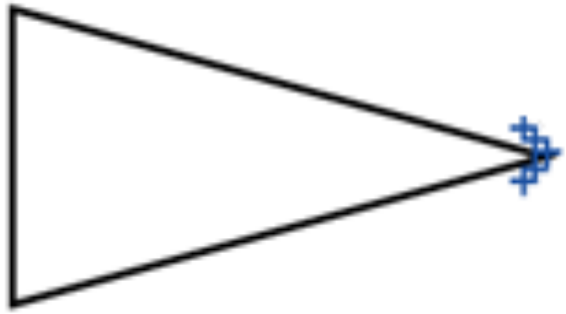
$E = \rho / \varepsilon_o$



r ↑

ρ ↓

E ↓



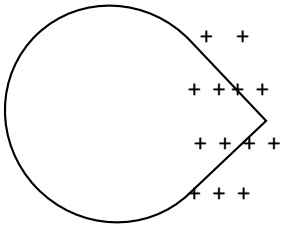
r ↓

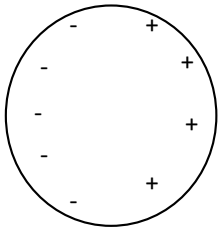
ρ ↑

E ↑

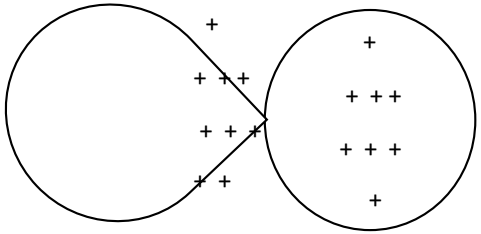
Point Discharge or Corona Discharge

1. Formation of Charged Particles





Air molecules or dust particles

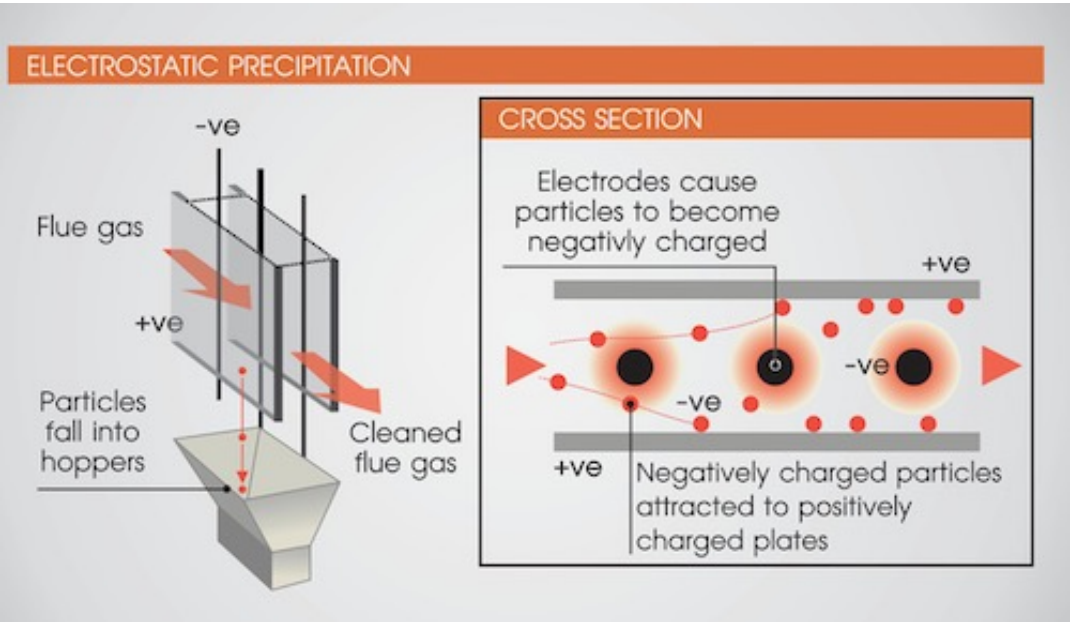


- E is very high at sharp edges.
- High E field polarizes air molecules and dust particles around it.
- The oppositely charged ends attract each other.
- Upon contact, they become positively charged objects and repel each other.
- This repelling of positively charged molecules (ionized molecules) generates an electric wind. When this continues to happen, the charges at sharp point will be lost to the surrounding rapidly. So, this process is called Corona Discharge (point discharge).
- Corona Discharge has many practical applications, e.g.:
- Lightning conductors.
- Van de Graaff generator (to obtain very high voltage – approximately 5 MV)

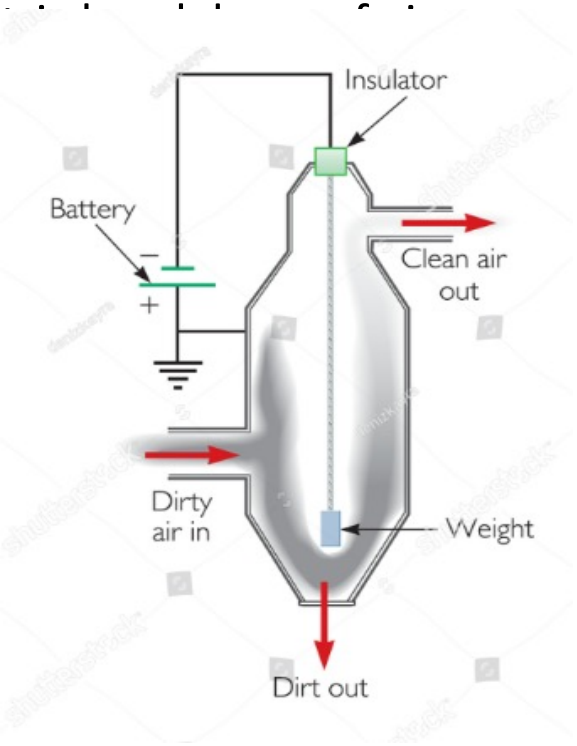
Technologically Important Applications

1. Electrostatic Precipitator

- An electric field is created between the wall and the corona wire. The electrostatic field close to the wire is large enough to produce an elec



https://oransi.com/blogs/blog/electrostatic-precipitator?srsltid=AfmBOopxBB8zNFJzFaJTDGyptiVvzu5fTG5MFwZKm0FP9jPr_yAG33WS



(https://share.google/AGVEBvxkEIRhnffPH)

- In the process of electric breakdown of air, accelerated electron hit the molecules.
- e.g.: $e^- + O_2 \rightarrow 2\ O^-$
- O^- ions are also accelerated. The small dust particles become charged by capturing ions.

As the dirty gas passes through this corona, free electrons and gas ions collide with the incoming dust or ash particles. This transfers a **strong negative electrical charge** to the pollutants.

collector plates that carry a positive charge. Driven by electrostatic force, the negatively charged particles migrate sideways—out of the horizontal gas stream—and tightly stick to the positively charged plates. The clean gas then safely exits through a stack into the atmosphere

- 99% cleaned air can be taken out.

- **Other Examples of Applications**
- 1. The electrostatic coating of spray paint.
- 2. Crop spraying from air.
- 3. Electrostatic separation of granular mixtures, e.g.:
  - (i) Cleaning of rice
  - (ii) Removal of weeds from wheat
  - (iii) Removal of rock particles from minerals

In Class Problems
PHY1203
Electrical and Magnetic Phenomena

August 14, 2026

Problem 2

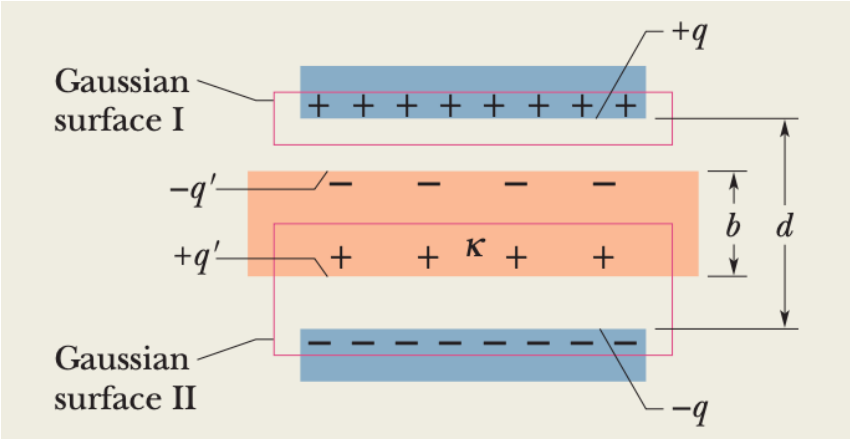


Figure 2. A parallel-plate capacitor containing a dielectric slab that only partially fills the space between the plates.

Figure 2 shows a parallel-plate capacitor of plate area A and plate separation d. A potential difference V_0 is applied between the plates by connecting a battery between them. The battery is then disconnected, and a dielectric slab of thickness b and dielectric constant K is placed between the plates as shown. Assume A =115 cm^2 , d = 1.24 cm, V_0 = 85.5 V, b = 0.780 cm, and K = 2.61.

(a) What is the capacitance C_0 before the dielectric slab is inserted?

(b) What free charge appears on the plates?

(c) What is the electric field $\boldsymbol{E_0}$ in the gaps between the plates and the dielectric slab?

(d) What is the electric field $\boldsymbol{E_1}$ in the dielectric slab

(e) What is the potential difference V between the plates after the slab has been introduced?

(f) What is the capacitance with the slab in place

Reference: Fundamental of Physics Haliday and Resnik, 10th Edition By Jearl Walker, pages. 736-737